

Consumer Search Behavior and Pricing

“When Do Consumers Search?”

Matt Lewis

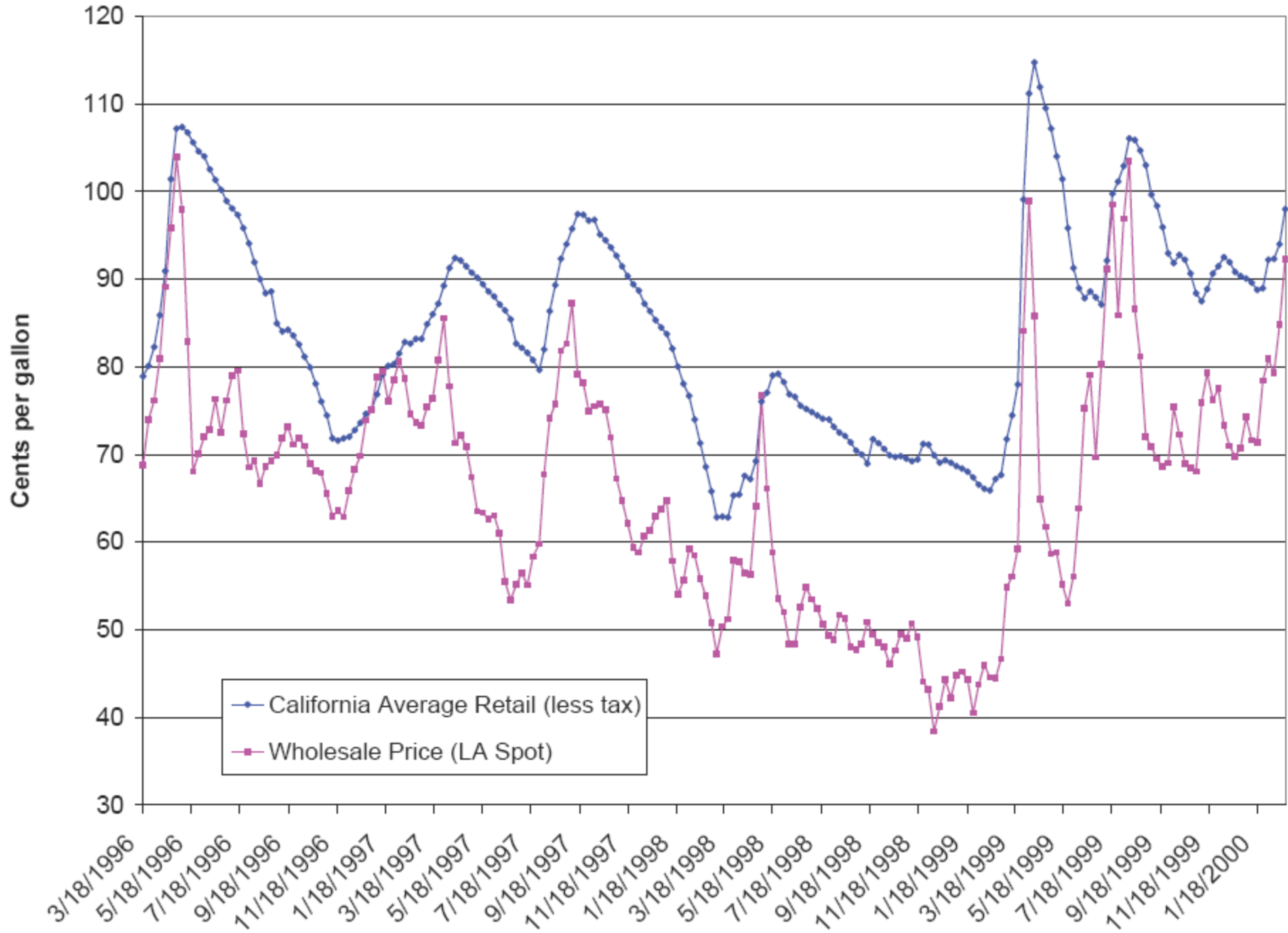
Journal of Industrial Economics,
2011

(Matt Lewis provided some of these slides)

What We Know About Search

- Literature weak on direct evidence of search
 - Brown & Goolsbee an exception but even there evidence is indirect
- Mostly infers search (or search costs) from data on prices

MOTIVATION: Weekly California Gasoline Prices 1996 - 2000



Broad Questions

- Do consumers search more when price levels are higher (but dispersion not necessarily high)?
 - Consumer benefit of search = $f(\text{dispersion})$
- Do consumers search more when dispersion is higher?
 - Can't directly measure, but can get at indirectly

Big Picture of Search & Gasoline Pricing

- Demand side:
 - Price movements (levels and variance) can potentially drive search behavior
- Supply side:
 - Search intensity affects local market power and thus pricing
- Instrumental variable:
 - Wholesale price of gasoline

Consumer information on price distribution

- Most models assume that consumers know the distribution of prices
- But what if they don't?
- Sketch of an alternative search model where consumers have incomplete information on price distribution

Search & asymmetric price adjustment

- Lewis (2005) – consumers search with expectations based on recent prices
- When prices increase:
 - most consumers encounter higher than expected prices
 - perceived value of search is higher
 - more consumers search, increasing competition and leading to more uniform pricing

Search & asymmetric price adjustment

- When prices decrease:
 - most consumers encounter lower than expected prices
 - perceived value of search is lower
 - fewer consumers search, decreasing competition and leading to less uniform pricing
- Consumers searching more when it is less beneficial

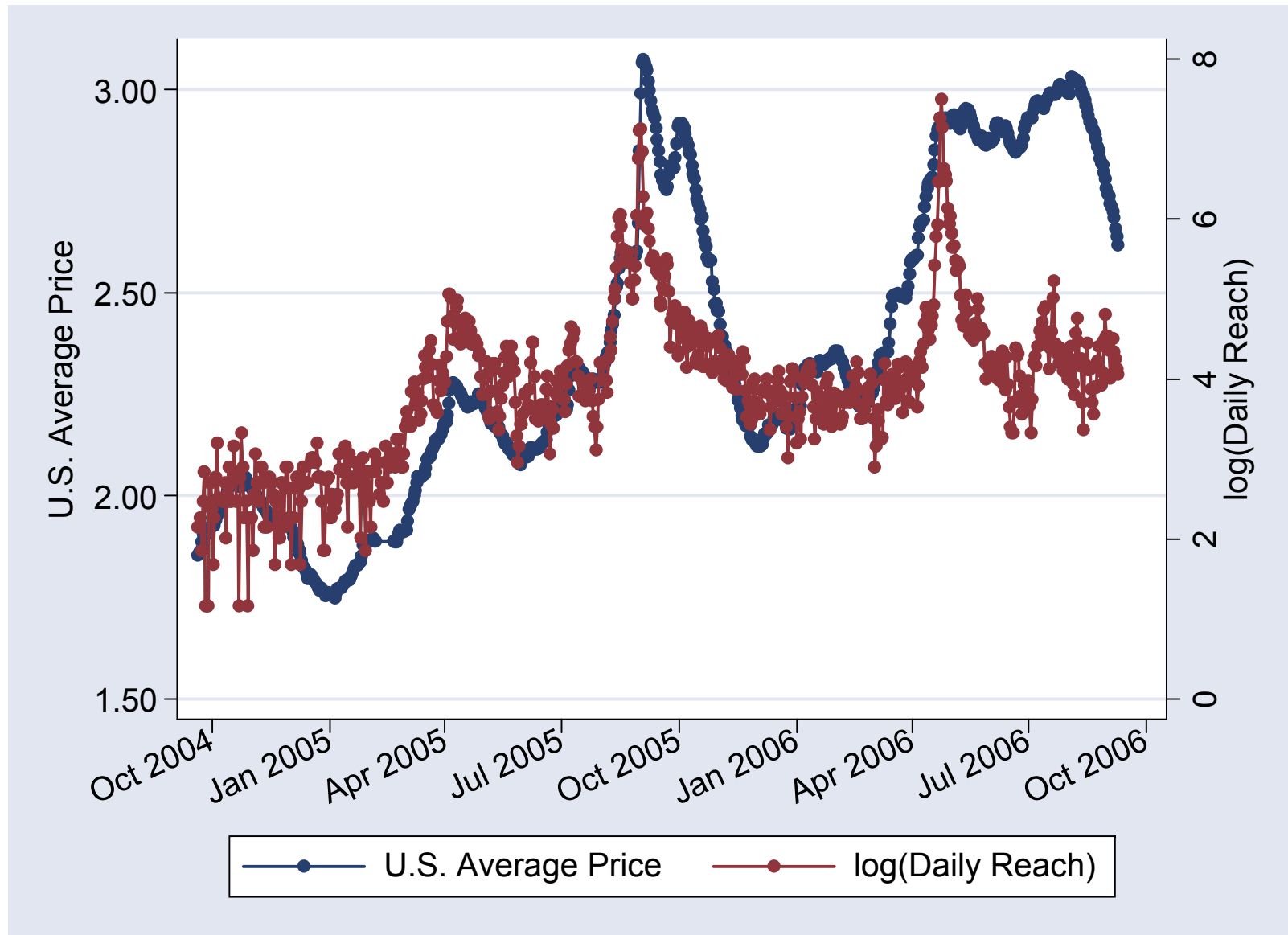
Data

- Read section III on Data!
- Doesn't just "take the data and run with it"
- Lewis thinks very carefully about accuracy and whether biases in reported search could be associated with directions of price changes
- This careful attention will be appreciated by (some) referees!

What the raw data say

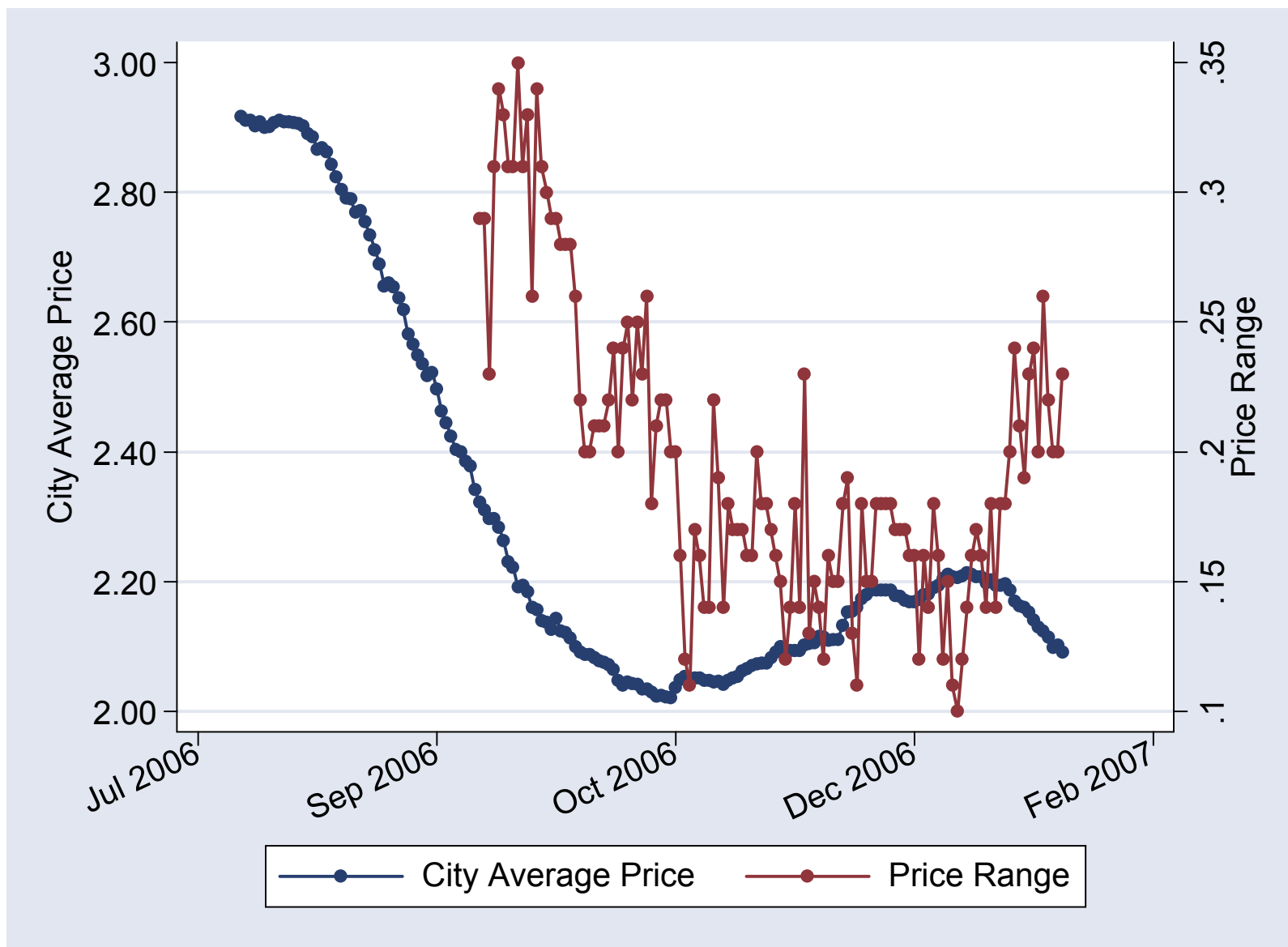
- search and dispersion change significantly over time and are related to overall gas price movements...

Gas prices (levels) and search



Gas Prices: Levels vs. Dispersion

Dallas average gasoline price and dispersion



GasBuddy.com

- Collects and reports station prices
- Main site is portal to state and city pages
- Most popular site for gasoline prices
- A good place to search



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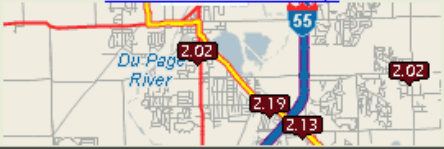
GasBuddy.com
GasBuddy.com can help you find cheap gas prices in your city. It is a network of more than 179+ gas price information web sites that helps you find low gasoline prices. All web sites are operated by GasBuddy and has the most comprehensive listings of gas prices anywhere.

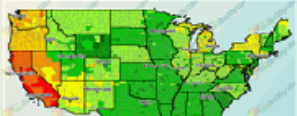
Gasoline prices change frequently and may vary by as much as 20 percent within only a few blocks it is important to be able locate the service station with the lowest priced fuel. GasBuddy web sites allow motorists to share information about low priced fuel with others as well as target the lowest priced stations to save money when filling up at the pumps!

Regular Gasoline Average Prices

	USA	Canada
Today	2.831	107.844
Yesterday	2.826	107.226
One Week ago	2.760	105.448
One Month ago	2.568	102.873
One Year ago	2.714	105.744
Current Trend		

[See the Historical Price Charts](#)

Map Gas Prices - USA Only
Find the lowest price of gas using a map.
[Click here to use the map!](#)


Gas Price Temperature Maps


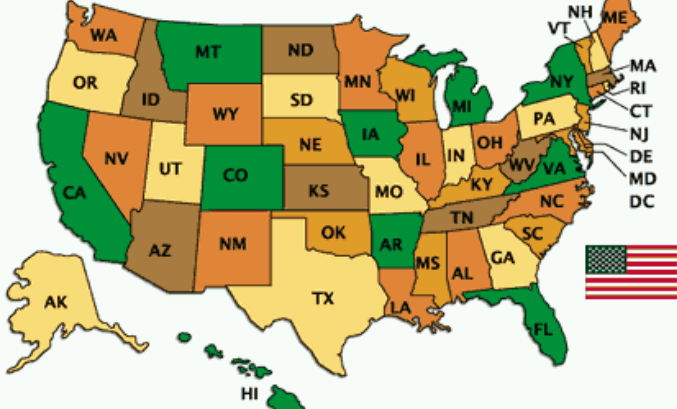
Gas Stations Hate Us
5 tips to save 20% at any gas station. Never pay full price!

Gas Prices Worldwide
12 cents per gallon in Venezuela How much could we be paying?

Ads by Google

Search by US Zip Code or City, State

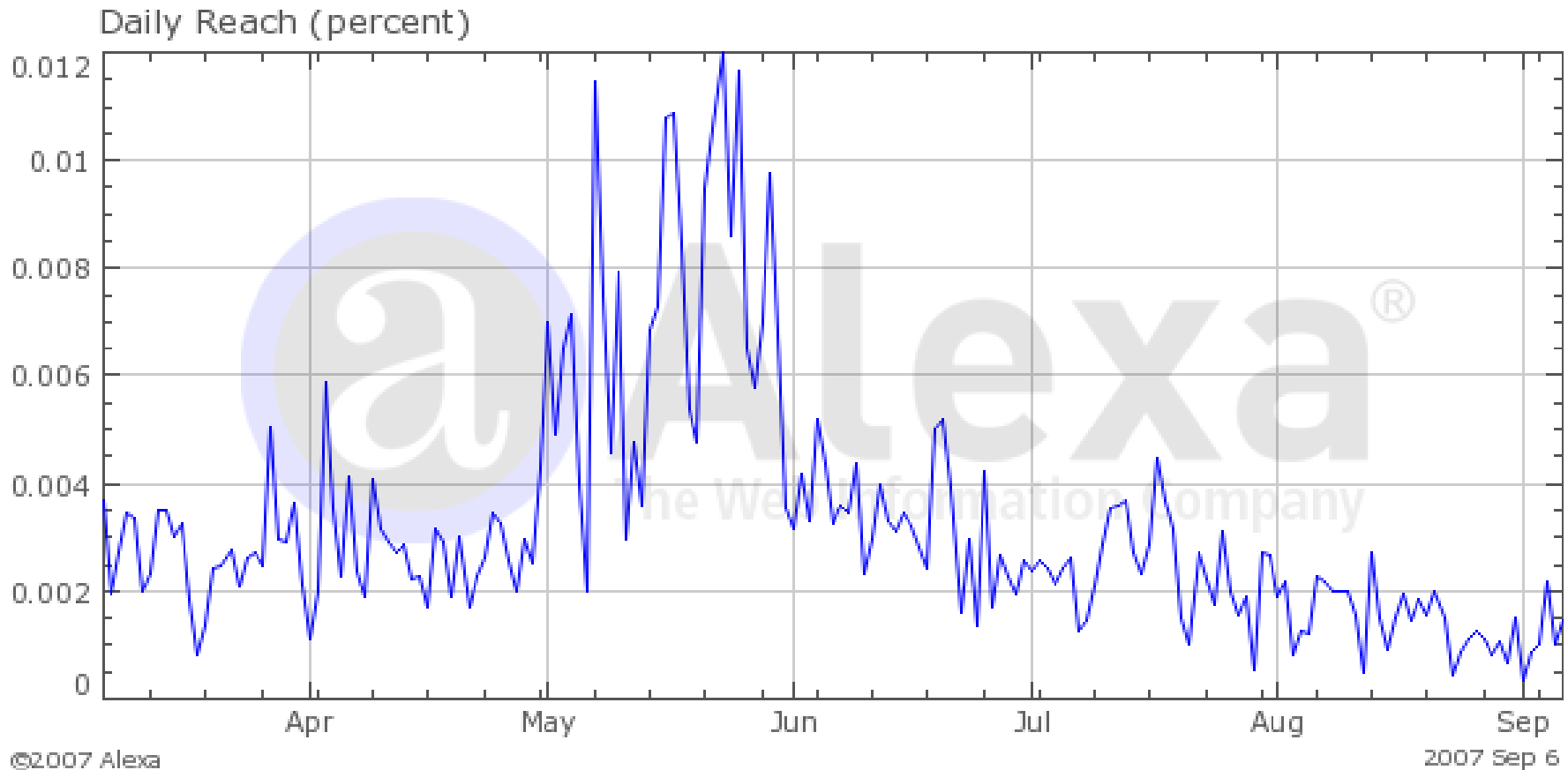
[Gas Prices by State/Province](#)

SELECT A LOCATION TO VIEW TODAY'S LOW AND HIGH GAS PRICES


How often is GasBuddy visited?

- Daily traffic statistics from Alexa.com
- Alexa toolbar users
- Traffic information for top 100,000 sites
- Historical traffic data available graphically
- Enguage digitizer program used to extract numerical traffic data from Alexa graphs
 - <http://digitizer.sourceforge.net/>

recent Alexa traffic chart for GasBuddy



Reach = % of all internet users visiting the site (ie. 0.01 = 10 out of every 100,000)

GasBuddy traffic data

- *Daily Reach* collected for 2 year period (Sept. 2004 – Sept. 2006)
- GasBuddy.com was consistently in the top 100,000 sites.
- Almost all city sites are not in top 100,000
- Daily traffic data matched with daily U.S. average retail gasoline price reported by GasBuddy.com

empirical analysis

Question 1: Does search respond asymmetrically to positive and negative price changes?

Strategy: Estimate a distributed lag model of search on past price changes

definitions

- Our price series, $\{p_t\}$, consists of a national average of prices for each day t from GasBuddy.com
- Define: $\Delta p_t \equiv p_t - p_{t-1}$

and

$$\xi_t \equiv \begin{cases} 1 & \text{if } \Delta p_t > 0, \text{ and} \\ 0 & \text{otherwise.} \end{cases}$$

specification

$$\begin{aligned}\ln(Reach)_t = & \alpha_0 + \alpha_1 \sum_{k=0}^4 \Delta p_{t-k} \xi_{t-k} + \alpha_2 \sum_{k=0}^4 \Delta p_{t-k} (1 - \xi_{t-k}) \\ & + \alpha_3 \sum_{k=5}^{19} \Delta p_{t-k} \xi_{t-k} + \alpha_4 \sum_{k=5}^{19} \Delta p_{t-k} (1 - \xi_{t-k}) \\ & + \alpha_5 \sum_{k=20}^{49} \Delta p_{t-k} \xi_{t-k} + \alpha_6 \sum_{k=20}^{49} \Delta p_{t-k} (1 - \xi_{t-k}) \\ & + \alpha_7 \text{date} + \alpha_8 \text{date}^2 + \text{day-of-week indicator variables} + \epsilon_t\end{aligned}$$

- IV with wholesale price changes
- Newey-West standard error estimates are used to control for serial correlation

Dependant Variable: $\ln(\text{Reach})$

Variable	Coefficients (Newey-West Standard Errors)			
	OLS	Instrumental Variables		
	(1)	(2)	(3)	(4)
α_0 (intercept)	2.09*** (0.28)	2.13*** (0.28)	2.17*** (0.20)	1.15* (0.62)
α_1 price change, +, most recent 5 days)	3.13*** (0.71)	2.20** (1.06)	1.72* (1.02)	2.11*** (0.40)
α_2 (price change, -, most recent 5 days)	-1.19 (1.19)	2.33 (2.56)	2.70 (2.51)	2.06 (1.45)
α_3 (price change, +, days 5-19)	3.11*** (0.71)	3.13*** (0.71)	2.85*** (0.56)	1.69*** (0.43)
α_4 (price change, -, days 5-19)	-1.10 (0.84)	-1.02 (0.85)	-1.06 (0.71)	-0.50 (0.66)
α_5 (price change, +, days 20-49)	1.11** (0.49)	1.20** (0.50)	0.93** (0.36)	0.48 (0.33)
α_6 (price change, -, days 20-49)	1.01** (0.51)	0.80 (0.54)	0.86** (0.37)	0.96 (0.64)
α_7 ($\ln(\text{Reach})$ for fuelgaugeport.com)			0.26*** (0.05)	
Website for $\ln(\text{Reach})$	GasBuddy	GasBuddy	GasBuddy	GasPriceWatch
Number of Observations	667	667	667	667

interpretation of search results

- Due to log-linear functional form:
 - %change in $Y = 100(e^{\beta}-1)$
 - a 10 cent **increase** in price occurring within the last 5 days is associated with a **24% increase** in search
 - a 10 cent **decrease** in price during the last 5 days is associated with **no significant change** in search (20.5% increase)

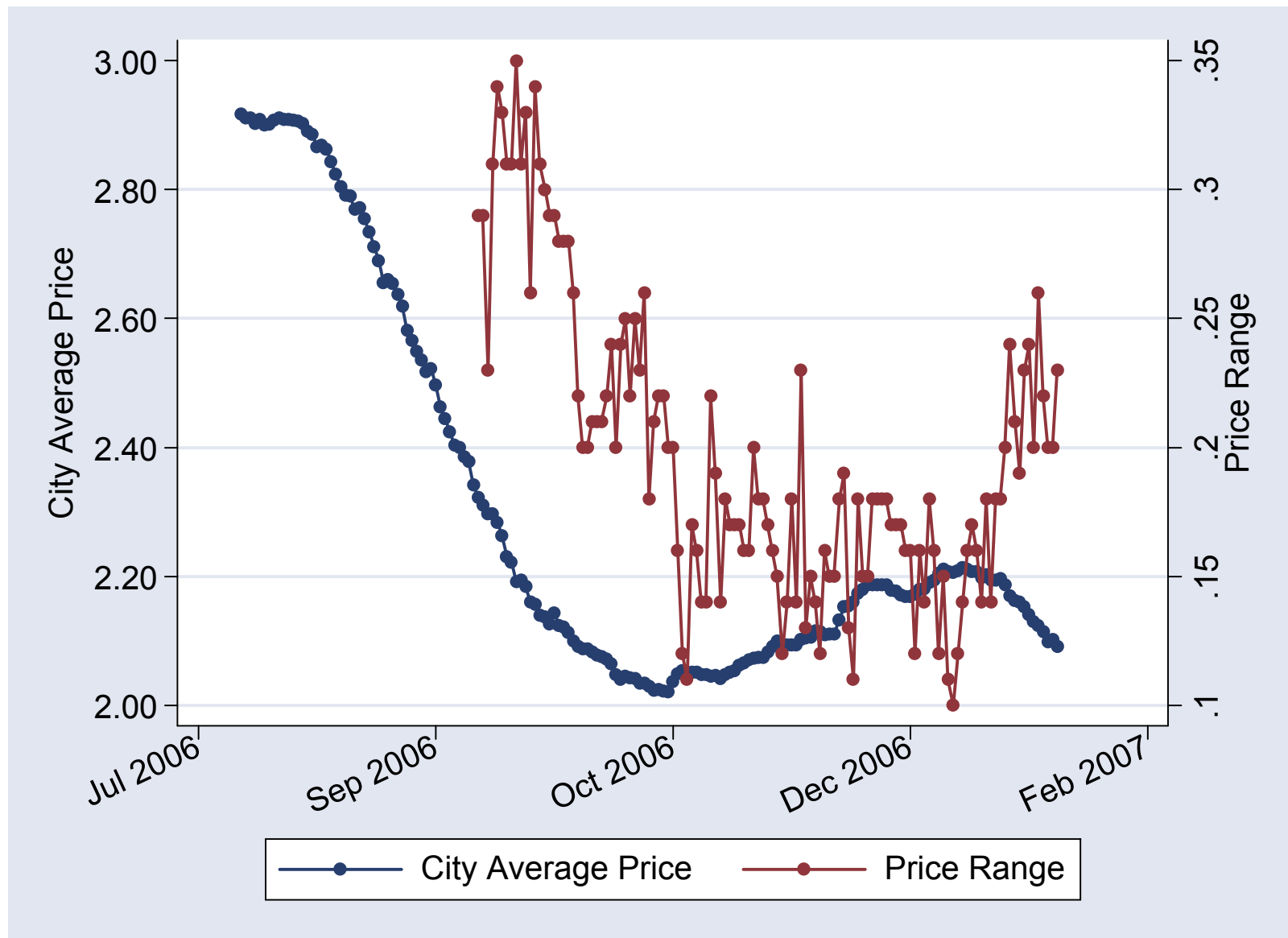
Next: Price Dispersion

- Search is greater following a price increase
 - Does search lead to a reduction in price dispersion?
 - Does search occur because price dispersion is greater?
- Price volatility in general should contribute to increased dispersion.
- If consumers search more when prices rise, then rising prices could have a much larger negative effect on dispersion than falling prices.

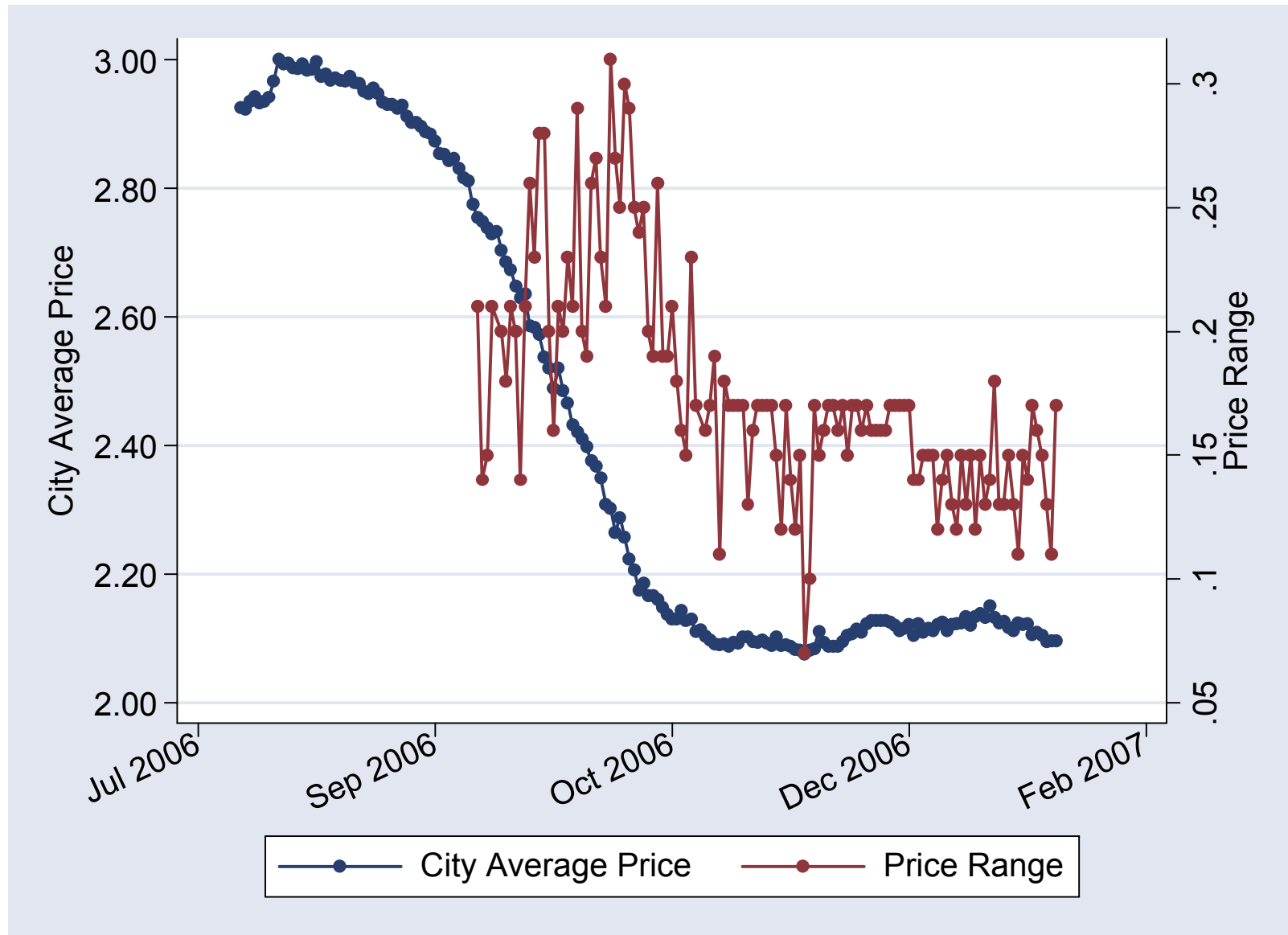
measuring price dispersion

- Dispersion within local markets measured by range of prices observed
- 2 measures of range:
 - Absolute range
 - Difference between the median of the high prices reported and the median low price
- alternative measures (i.e. variance) not possible with available data

Dallas gas prices and price dispersion



Denver gas prices and price dispersion



empirical analysis

Question 2: Does dispersion respond asymmetrically to positive and negative price changes?

Strategy: Estimate a distributed lag model of price dispersion on past price changes

specification

$$\begin{aligned}\ln(Range)_{ct} = & \beta_1 \sum_{k=0}^4 \Delta p_{t-k} \xi_{t-k} + \beta_2 \sum_{k=0}^4 \Delta p_{t-k} (1 - \xi_{t-k}) + \beta_3 \sum_{k=5}^{19} \Delta p_{t-k} \xi_{t-k} \\ & + \beta_4 \sum_{k=5}^{19} \Delta p_{t-k} (1 - \xi_{t-k}) + \beta_5 \sum_{k=20}^{49} \Delta p_{t-k} \xi_{t-k} + \beta_6 \sum_{k=20}^{49} \Delta p_{t-k} (1 - \xi_{t-k}) \\ & + \sum_{1}^6 \text{day of week}_t + \sum_{1}^C \text{city}_c + \epsilon_{ct}\end{aligned}$$

- 49-103 cities, depending on measure of range employed
- City and day-of-week fixed effects
- Note: prices are now city specific averages

		Specification			
Sample:		49 cities	City/date pairs with 30 prices		full sample
Dependent Variable:		median range	$p^{\max} - p^{\min}$	median range	$p^{\max} - p^{\min}$
Greater dispersion when prices are falling	Coefficients (Newey-West standard errors in parenthesis)				
	β_1 (price change +, most recent 4 days)	0.30 (0.173)	0.11 (0.119)	0.25* (0.147)	0.10 (0.129)
	β_2 (price change, -, most recent 4 days)	-2.41*** (0.222)	-1.81*** (0.153)	-2.31*** (0.178)	-1.25*** (0.183)
	β_3 (price change, +, days 5-20)	-0.13 (0.101)	-0.27*** (0.089)	-0.13 (0.097)	-0.55*** (0.100)
	β_4 (price change, -, days 5-20)	-1.02*** (0.094)	-1.09*** (0.068)	-1.04*** (0.072)	-1.12*** (0.080)
	β_5 (price change, +, days 20-50)	-0.01 (0.057)	-0.08* (0.047)	-0.08 (0.053)	-0.14*** (0.052)
	β_6 (price change, -, days 20-50)	-0.14*** (0.037)	-0.13*** (0.030)	-0.13*** (0.032)	-0.12*** (0.032)

conclusions

- A first look at actual search behavior reveals:
 - search increases most when dispersion decreases
 - suggests search may respond to price changes directly rather than to dispersion
 - price changes stimulate search asymmetrically
- Evidence supports consumer search as potential explanation for asymmetric retail gasoline price adjustment